

COL724/7524 Advanced Computer Networks

Assignment 1 – Part C1

Static Route Policy

1 Objective

Part C1 demonstrates the use of a static route to explicitly control the forwarding path of packets in an inter-AS network.

The topology contains four autonomous systems connected in a chain:

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$$

In addition, there is a direct link between AS1 and AS4:

$$AS1 \rightarrow AS4$$

Thus, two possible paths exist between AS1 and AS4:

- P_1 : $AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$
- P_2 : $AS1 \rightarrow AS4$

A static route is configured at AS1 so that traffic destined for the AS4 network is explicitly forwarded through AS2. Therefore, the forward traffic follows the longer path despite the availability of the direct AS1–AS4 link.

2 Network Topology

The simulated network consists of four routers representing the four autonomous systems. The links are arranged as follows:

$$AS1 - AS2 - AS3 - AS4$$

with an additional direct connection:

$$AS1 - AS4$$

The static routing policy is applied at AS1 to force packets towards the AS4 destination through AS2.

3 Static Route Policy

The main purpose of Part C1 is to demonstrate explicit forwarding using a manually configured static route.

For traffic originating from AS1 and destined for AS4, the intended forwarding path is:

$$\boxed{AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4}$$

Although the direct AS1–AS4 connection is available, the static route at AS1 causes the packet to be sent towards AS2.

This experiment therefore demonstrates that a static forwarding rule can be used to control the path taken by packets independently of the direct path available in the topology.

4 Simulation Setup

The simulation was implemented using ns-3. The experiment uses UDP traffic between the two endpoint networks.

The forward flow is:

$$192.168.1.1 \rightarrow 192.168.3.2$$

and the reverse flow is:

$$192.168.4.2 \rightarrow 192.168.1.1$$

The simulation generates eight packets in each direction.

5 Console Output

The console output confirms that packets are transmitted and received successfully during the simulation.

The client sends packets towards the destination, while the server receives the packets and sends the corresponding response traffic.

```

soham-malvadar@Malvadar-OMEN:~/ns-3.48$ ./ns3 run scratch/bgp_partC1.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
Starting inter-AS routing animation with AS1-AS2-AS3-AS4 and AS1-AS4 links...
At time +2s client sent 512 bytes to 192.168.3.2 port 8080
At time +2.01181s server received 512 bytes from 192.168.1.1 port 49153
At time +2.01181s server sent 512 bytes to 192.168.1.1 port 49153
At time +2.0132s client received 512 bytes from 192.168.4.2 port 8080
At time +2.75s client sent 512 bytes to 192.168.3.2 port 8080
At time +2.76181s server received 512 bytes from 192.168.1.1 port 49153
At time +2.76181s server sent 512 bytes to 192.168.1.1 port 49153
At time +2.7632s client received 512 bytes from 192.168.4.2 port 8080
At time +3.5s client sent 512 bytes to 192.168.3.2 port 8080
At time +3.51181s server received 512 bytes from 192.168.1.1 port 49153
At time +3.51181s server sent 512 bytes to 192.168.1.1 port 49153
At time +3.5132s client received 512 bytes from 192.168.4.2 port 8080
At time +4.25s client sent 512 bytes to 192.168.3.2 port 8080
At time +4.26181s server received 512 bytes from 192.168.1.1 port 49153
At time +4.26181s server sent 512 bytes to 192.168.1.1 port 49153
At time +4.2632s client received 512 bytes from 192.168.4.2 port 8080
At time +5s client sent 512 bytes to 192.168.3.2 port 8080
At time +5.01181s server received 512 bytes from 192.168.1.1 port 49153
At time +5.01181s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.0132s client received 512 bytes from 192.168.4.2 port 8080
At time +5.75s client sent 512 bytes to 192.168.3.2 port 8080
At time +5.76181s server received 512 bytes from 192.168.1.1 port 49153
At time +5.76181s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.7632s client received 512 bytes from 192.168.4.2 port 8080
At time +6.5s client sent 512 bytes to 192.168.3.2 port 8080
At time +6.51181s server received 512 bytes from 192.168.1.1 port 49153
At time +6.51181s server sent 512 bytes to 192.168.1.1 port 49153
At time +6.5132s client received 512 bytes from 192.168.4.2 port 8080
At time +7.25s client sent 512 bytes to 192.168.3.2 port 8080
At time +7.26181s server received 512 bytes from 192.168.1.1 port 49153
At time +7.26181s server sent 512 bytes to 192.168.1.1 port 49153
At time +7.2632s client received 512 bytes from 192.168.4.2 port 8080
soham-malvadar@Malvadar-OMEN:~/ns-3.48$ 

```

Figure 1: Console output of the Part C1 static routing simulation.

From the output, packets can be observed being exchanged between the following addresses:

- 192.168.1.1
- 192.168.3.2
- 192.168.4.2

The successful send and receive events indicate that the configured static route is operational.

6 FlowMonitor Results

The FlowMonitor output contains two flows.

Table 1: FlowMonitor packet statistics

Direction	Tx Packets	Rx Packets	Lost Packets
AS1 → AS4	8	8	0
AS4 → AS1	8	8	0

Both flows successfully deliver all eight packets. Therefore, the simulation experiences no packet loss.

The packet delivery rate is:

$$\text{Delivery Rate} = \frac{8}{8} \times 100 = 100\%$$

and the packet loss is:

$$\boxed{0\%}$$

7 Forward Path Analysis

For the forward flow from AS1 to AS4, the FlowMonitor output reports:

$$\text{timesForwarded} = 16$$

There are eight transmitted packets. Since the packets are forwarded through the two intermediate routers, AS2 and AS3, the expected number of forwarding events is:

$$8 \times 2 = 16$$

which matches the FlowMonitor result.

Therefore, the forward traffic follows:

$$\boxed{AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4}$$

This provides evidence that the static route successfully forces the forward traffic through the intermediate autonomous systems instead of using the direct AS1–AS4 link.

8 Forward Average Delay

For the forward flow, FlowMonitor reports:

$$\text{delaySum} = 9.44533 \times 10^7 \text{ ns}$$

and:

$$\text{rxPackets} = 8$$

Therefore, the average delay is:

$$D_{\text{forward}} = \frac{9.44533 \times 10^7}{8}$$

$$D_{\text{forward}} = 1.18066625 \times 10^7 \text{ ns}$$

Converting to milliseconds:

$$D_{\text{forward}} = 11.8066625 \text{ ms}$$

Hence,

$$\boxed{D_{\text{forward}} \approx 11.8067 \text{ ms}}$$

The relatively large delay is expected because the packets traverse the three-hop AS1–AS2–AS3–AS4 path.

9 Reverse Path Analysis

The reverse flow corresponds to traffic from AS4 towards AS1:

$$192.168.4.2 \rightarrow 192.168.1.1$$

For this flow, FlowMonitor reports:

$$\text{timesForwarded} = 0$$

This indicates that the reverse packets are not forwarded through the intermediate routers.

Therefore, the reverse path is:

$$\boxed{AS4 \rightarrow AS1}$$

using the direct AS4–AS1 connection.

This produces an asymmetric routing behavior: the forward traffic is forced through AS2 and AS3, while the reverse traffic uses the direct link.

10 Reverse Average Delay

For the reverse flow, FlowMonitor reports:

$$\text{delaySum} = 1.11535 \times 10^7 \text{ ns}$$

and:

$$\text{rxPackets} = 8$$

Thus,

$$D_{\text{reverse}} = \frac{1.11535 \times 10^7}{8}$$

$$D_{\text{reverse}} = 1.3941875 \times 10^6 \text{ ns}$$

or:

$$D_{\text{reverse}} \approx 1.3942 \text{ ms}$$

The much smaller delay compared with the forward direction is consistent with the reverse traffic using the direct AS4–AS1 link.

11 Forward and Reverse Comparison

The observed routing behavior can be summarized as follows.

Table 2: Comparison of forward and reverse traffic

Direction	Path	Average Delay	Times Forwarded
AS1 → AS4	AS1–AS2–AS3–AS4	11.8067 ms	16
AS4 → AS1	AS4–AS1	1.3942 ms	0

The forward direction has a significantly larger delay because the static route forces packets through two intermediate routers.

The reverse direction uses the direct link and therefore has a much smaller delay.

12 NetAnim Visualization

The simulation was also visualized using NetAnim. The animation shows the four-router topology and the movement of packets during the simulation.

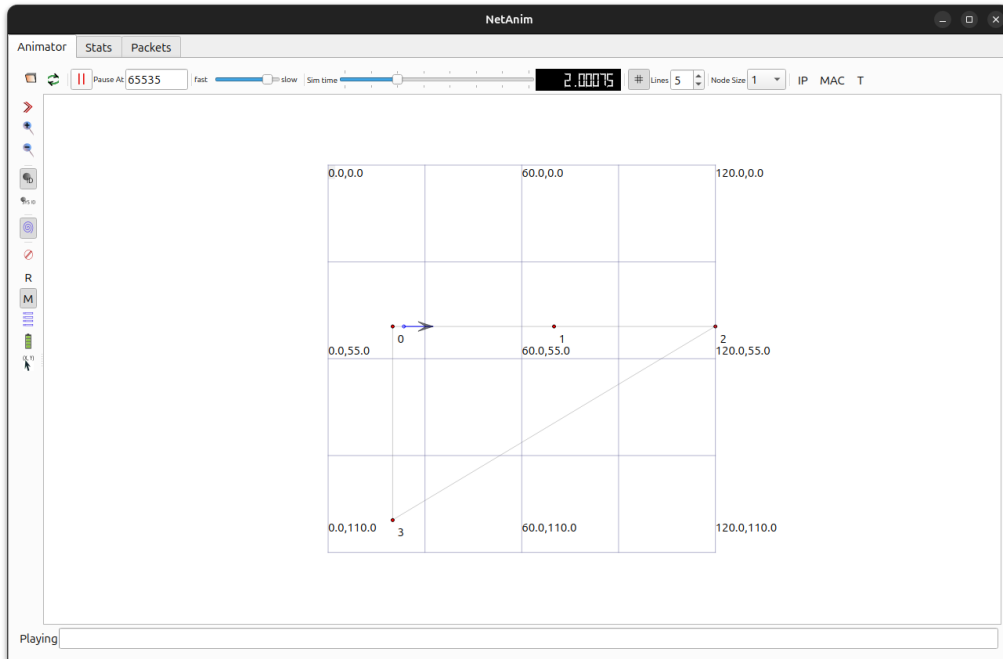


Figure 2: NetAnim view showing the Part C1 topology and packet movement.

Figure 2 shows the four routers and the direct AS1–AS4 connection. Packet movement can be observed during the simulation.

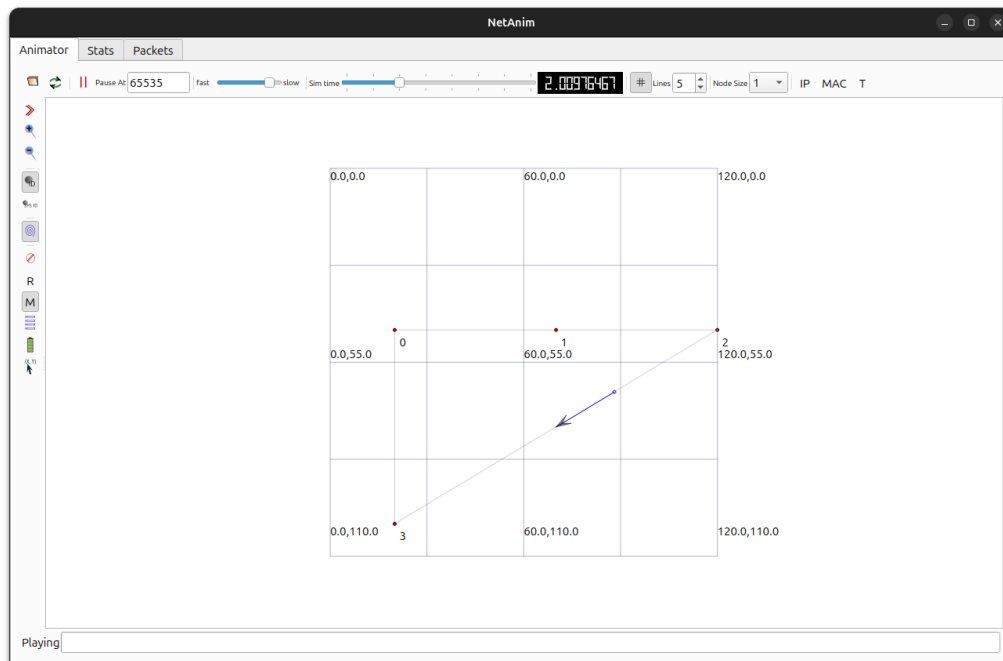


Figure 3: NetAnim frame showing packet movement on the simulated network.

The animation provides visual confirmation of packet activity in the network and complements the forwarding statistics obtained from FlowMonitor.

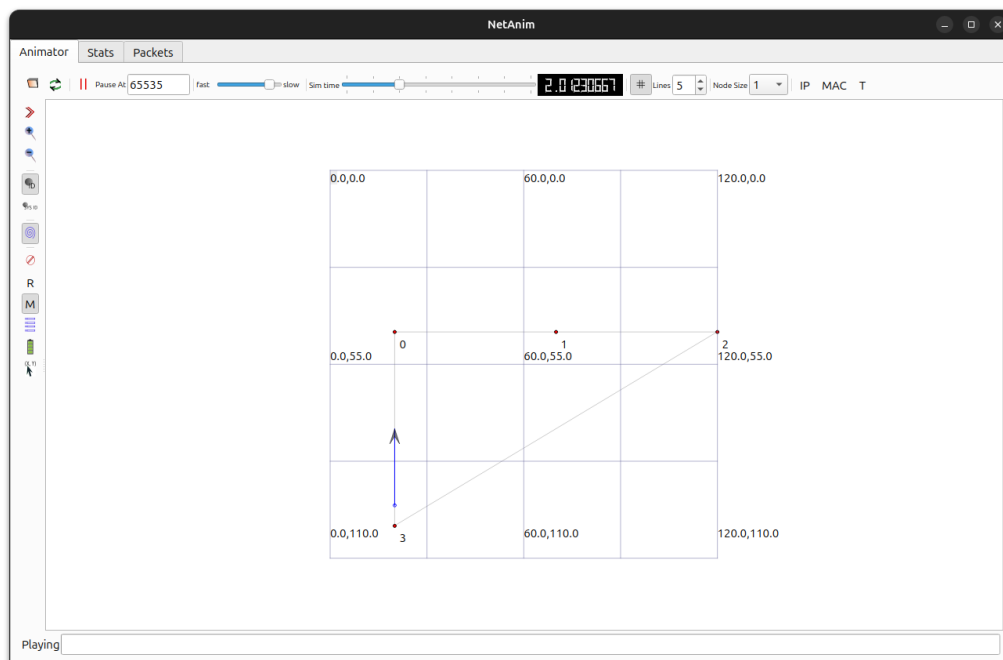


Figure 4: NetAnim frame showing packet transmission between the routers.

13 Observations

The main observations from the Part C1 experiment are:

1. A static route can explicitly control the forwarding path of packets.
2. The forward AS1-to-AS4 traffic is forced through:

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$$

3. The forward flow has 16 forwarding events for 8 packets, confirming forwarding through the two intermediate routers.
4. The reverse flow has zero forwarding events and therefore uses the direct AS4–AS1 connection.
5. The average forward delay is approximately 11.8067 ms.
6. The average reverse delay is approximately 1.3942 ms.
7. Both directions achieve 100% packet delivery with 0% packet loss.
8. The experiment demonstrates asymmetric routing caused by the static forwarding policy.

14 Conclusion

Part C1 demonstrates how static routing can be used to explicitly control packet forwarding in an inter-AS network.

Even though a direct AS1–AS4 connection is available, the static route configured at AS1 forces traffic towards the AS4 network through:

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$$

The FlowMonitor results support this behavior. The forward flow has 16 forwarding events for 8 packets, whereas the reverse flow has zero forwarding events.

The measured average delays are approximately:

$$11.8067 \text{ ms} \quad (\text{forward})$$

and

$$1.3942 \text{ ms} \quad (\text{reverse})$$

All packets are successfully delivered, resulting in a 100% delivery rate and 0% packet loss.

Overall, the experiment shows that static routing provides explicit path control and can produce asymmetric forwarding behavior even when multiple paths are available in the network.

COL724/7524 Advanced Computer Networks

Assignment 1

Part C2: Interface Metric Based Routing

Objective

The objective of this experiment is to demonstrate how modifying the interface metric can influence the route selected by the global routing mechanism in the four-router inter-AS topology.

The topology consists of four routers connected as

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$$

along with an additional direct link

$$AS1 \rightarrow AS4.$$

The purpose of this experiment is to make the route through AS2 and AS3 the preferred route while keeping the direct AS1–AS4 link available.

Thus, the desired forwarding path is

$$\boxed{AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4}.$$

Topology

The simulated topology contains four routers corresponding to the four autonomous systems. The links between the routers are:

- AS1–AS2
- AS2–AS3
- AS3–AS4
- AS1–AS4

The AS1–AS4 link provides a direct alternative path between the source and destination autonomous systems.

The experiment modifies the interface metric so that the global routing calculation prefers the path through AS2 and AS3.

Routing Policy

The routing policy implemented in this experiment is to prefer the multi-hop path

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$$

over the direct path

$AS1 \rightarrow AS4$.

The physical topology is not changed. The direct AS1–AS4 link remains connected and available.

Instead, the interface metric is modified. The resulting routing calculation therefore selects the path with the preferred metric.

This demonstrates that routing behaviour can be changed through interface metrics without removing or disabling an existing link.

Simulation Configuration

The simulation was executed using the C2 configuration with four routers. Traffic was generated from the client in AS1 towards the server in AS4.

The application traffic consists of packets of size 512 bytes. The simulation output shows repeated transmission and reception of packets between the client and server.

The relevant communication is of the form

Client (AS1) $\rightarrow$ Server (AS4)

with the return traffic travelling in the reverse direction.

Console Output

The simulation was successfully executed using the C2 configuration. The console output shows that the client repeatedly sends packets to the destination server and that the packets are successfully received.

```

Starting Inter-AS routing simulation with 4 routers (AS1-AS2-AS3-AS4)...
At time +2s client sent 512 bytes to 192.168.3.2 port 8080
At time +2.01181s server received 512 bytes from 192.168.1.1 port 49153
At time +2.01181s server sent 512 bytes to 192.168.1.1 port 49153
At time +2.01324s client received 512 bytes from 192.168.4.2 port 8080
At time +2.75s client sent 512 bytes to 192.168.3.2 port 8080
At time +2.76181s server received 512 bytes from 192.168.1.1 port 49153
At time +2.76181s server sent 512 bytes to 192.168.1.1 port 49153
At time +2.76324s client received 512 bytes from 192.168.4.2 port 8080
At time +3.5s client sent 512 bytes to 192.168.3.2 port 8080
At time +3.51181s server received 512 bytes from 192.168.1.1 port 49153
At time +3.51181s server sent 512 bytes to 192.168.1.1 port 49153
At time +3.51324s client received 512 bytes from 192.168.4.2 port 8080
At time +4.25s client sent 512 bytes to 192.168.3.2 port 8080
At time +4.26181s server received 512 bytes from 192.168.1.1 port 49153
At time +4.26181s server sent 512 bytes to 192.168.1.1 port 49153
At time +4.26324s client received 512 bytes from 192.168.4.2 port 8080
At time +5s client sent 512 bytes to 192.168.3.2 port 8080
At time +5.01181s server received 512 bytes from 192.168.1.1 port 49153
At time +5.01181s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.01324s client received 512 bytes from 192.168.4.2 port 8080
At time +5.75s client sent 512 bytes to 192.168.3.2 port 8080
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At time +5.76181s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.76324s client received 512 bytes from 192.168.4.2 port 8080
At time +6.5s client sent 512 bytes to 192.168.3.2 port 8080
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At time +7.26181s server received 512 bytes from 192.168.1.1 port 49153
At time +7.26181s server sent 512 bytes to 192.168.1.1 port 49153
At time +7.26324s client received 512 bytes from 192.168.4.2 port 8080
soham-malvadar@Malvadar-OMEN:~/ns-3.48$ 

```

Figure 1: Console output of the C2 simulation.

The console output contains repeated messages such as:

```

client sent 512 bytes to 192.168.3.2 port 8080
server received 512 bytes from 192.168.1.1
server sent 512 bytes to 192.168.1.1
client received 512 bytes from 192.168.4.2 port 8080

```

These messages indicate successful bidirectional communication between the source and destination.

Simulation Output 1

The first output screenshot shows the packet transmission behaviour in the simulated topology.

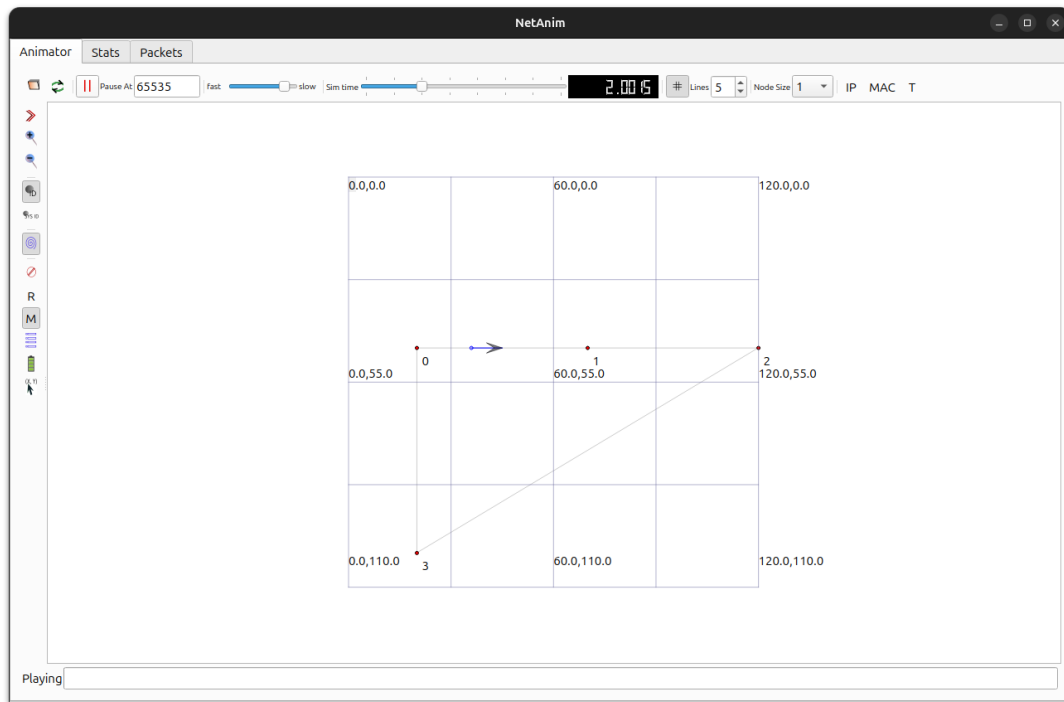


Figure 2: C2 simulation showing packet transmission.

The packet transmission confirms that traffic is generated from the source side and forwarded towards the destination network.

Simulation Output 2

The second output screenshot provides the corresponding NetAnim visualization of packet movement.

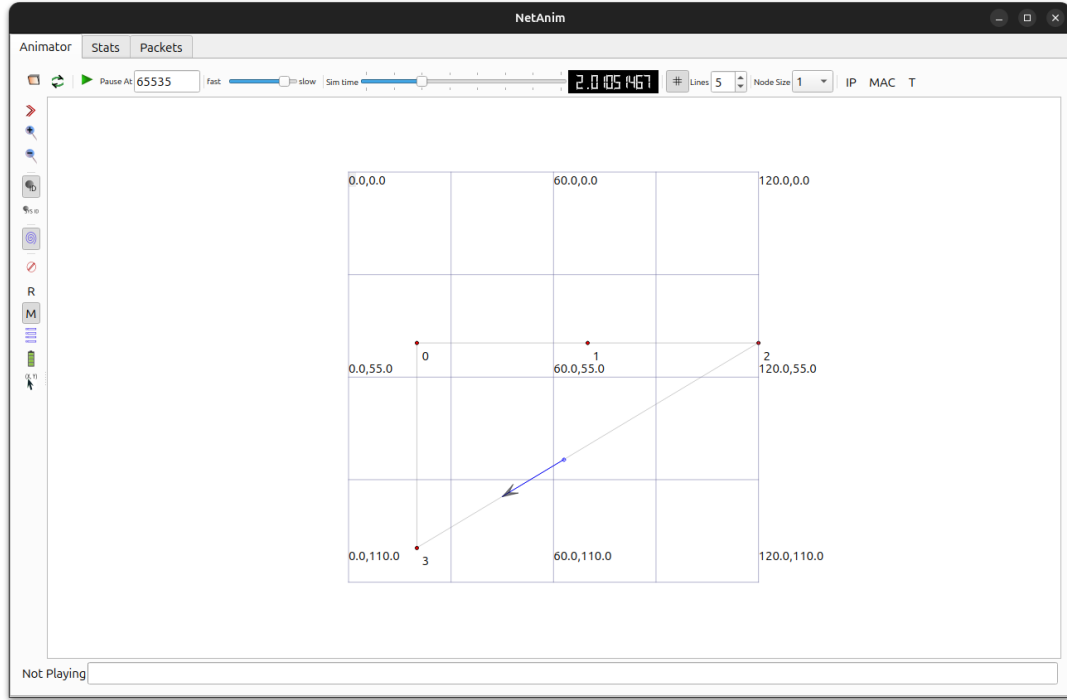


Figure 3: NetAnim visualization of packet forwarding in C2.

The animation shows packets travelling through the intermediate routers. The observed forwarding behaviour is consistent with the desired path

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4.$$

The direct AS1–AS4 link remains visible in the topology, but the observed forward traffic follows the route through AS2 and AS3.

Simulation Output 3

The third output screenshot provides additional evidence of the packet movement and successful delivery.

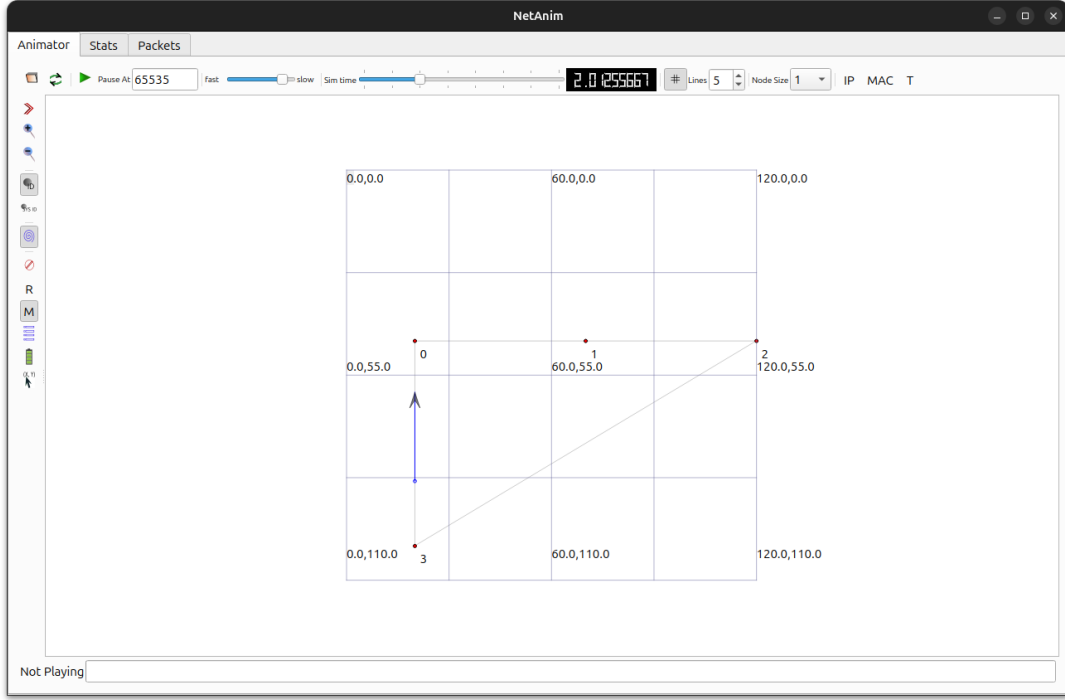


Figure 4: Additional NetAnim output showing packet forwarding.

The packet movement observed in the animation demonstrates that the modified metric affects the route selected by the global routing mechanism.

Forwarding Behaviour

The most important change introduced in C2 is the forwarding path.

Instead of using the direct link

$$AS1 \rightarrow AS4,$$

the packets are forwarded through the intermediate routers:

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4.$$

Therefore, the packet traverses two intermediate routers before reaching the destination.

The direct AS1–AS4 link is not removed from the topology. It remains available as an alternative path.

Hence, the experiment demonstrates that modifying the interface metric can affect route preference while preserving the underlying network topology.

FlowMonitor Results

The packet delivery behaviour can be evaluated using the FlowMonitor statistics obtained from the simulation.

Table 1: FlowMonitor results for the C2 experiment

Direction	Tx Packets	Rx Packets	Lost Packets
AS1 $\rightarrow$ AS4	8	8	0
AS4 $\rightarrow$ AS1	8	8	0

For the forward flow, the packet delivery ratio is

$$\text{PDR} = \frac{\text{Received Packets}}{\text{Transmitted Packets}} \times 100$$

and therefore

$$\text{PDR} = \frac{8}{8} \times 100 = 100\%.$$

The packet loss is

$$\text{Packet Loss} = \frac{8 - 8}{8} \times 100 = 0\%.$$

Thus, the metric modification changes the preferred route without causing packet loss in the observed flow.

Comparison with Direct Routing

The difference in forwarding behaviour can be summarized as follows.

Table 2: Comparison of direct routing and C2

Property	Direct Path	C2
Preferred path	AS1 $\rightarrow$ AS4	AS1 $\rightarrow$ AS2 $\rightarrow$ AS3 $\rightarrow$ AS4
Intermediate routers	0	2
AS1–AS4 link	Available	Available
Routing mechanism	Global routing	Global routing with modified interface metric
Packet loss	0%	0%

The important observation is that C2 does not remove the direct link. Instead, the interface metric influences the route selection process and causes the longer path to become preferred.

Comparison with Part C1

Part C1 and Part C2 implement the same overall routing policy, but they use different mechanisms to achieve it.

In Part C1, the desired path is explicitly configured using a static route. The forwarding decision is therefore manually specified by defining the appropriate next hop.

In Part C2, no explicit static route is used to force the path. Instead, the interface metric is modified so that the global routing mechanism prefers the path through AS2 and AS3.

The difference can be summarized as follows:

Table 3: Comparison of Part C1 and Part C2

Property	Part C1: Static Routing	Part C2: Interface Metric
Routing mechanism	Static route	Global routing with modified interface metric
Route selection	Explicitly specified by the configured next hop	Determined by the routing calculation based on interface costs
Preferred path	$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$	$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4$
Direct AS1–AS4 link	Remains available	Remains available
Manual route configuration	Required	Not required for selecting the preferred path
Purpose	Explicitly force a particular route	Influence route preference through metric values

Thus, although both C1 and C2 can result in the same forwarding path,

$$\boxed{AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4},$$

the method used to obtain this path is different. Part C1 directly specifies the route, whereas Part C2 changes the cost associated with interfaces and allows the global routing mechanism to select the preferred route.

Observation

The main observations from the experiment are:

1. The four-router topology remains unchanged.
2. The direct AS1–AS4 link remains available.
3. The interface metric is modified to influence route selection.
4. The preferred path becomes

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4.$$

5. Packets are successfully transmitted from the client to the server.
6. The return traffic is also successfully received.
7. The observed packet loss is zero.

Conclusion

The C2 experiment successfully demonstrates the effect of modifying interface metrics on global route selection.

Although the direct AS1–AS4 link is present in the topology, the modified interface metric causes the routing mechanism to prefer the path

$$\boxed{AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4}.$$

The console output confirms successful packet transmission and reception, while the NetAnim visualizations provide visual evidence of packet forwarding through the intermediate routers.

The experiment therefore shows that interface metrics can be used to influence routing decisions without explicitly installing a static route or changing the physical topology.